

CAMP COMMUNIQUE

PROMABRIDGE NATIONAL STEM INNOVATION CAMP 2026

Camp Theme: Zambia at 62: Building Self-Reliance through STEM Innovation

Slogan: STEM of Hope to Z Innovation





DATE AND LOCATION

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13th December To 20th December, 2026



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VISION, MISSION AND CAMP TARGETS

Vision



To raise a generation of Zambian innovators equipped with advanced STEM skills, entrepreneurial thinking, and global competitiveness.

Mission



To deliver immersive, hands-on STEM education through real-world engineering, technology, and innovation projects guided by industry experts and academic leaders.

Camp Targets



1. Minimum attendance: 1,000 learners
 2. Maximum attendance: 1,500 learners
- This scale ensures national impact while maintaining quality mentorship and effective project supervision.

Introduction

Promabridge Institute 2026

The Production And Manufacturing bridge National STEM Innovation Camp in partnership with Hope Empowerment foundation - is a hands-on learning experience designed to spark a love for mathematics, science, and technology while transforming learners from theory-focused students into practical problem-solvers. Through realworld projects and innovation, the camp equips young Zambians with skills to think, build, and lead the future.

Promabridge National STEM Camp 2026 (Rotational Learning Timetable)

HOSTS AND KEYNOTES



Host:
Blessing Banda



Co-Host:
Maureen Chibanga
(Chief Marketing Officer)



Keynote Speaker:
Pro. Clive Chirwa

Guests of Honour:

1. Minister of Science and Technology
2. Minister of General Education
3. Minister of Higher Education

From Monday to Friday, all technical halls operate simultaneously. Student campers are divided into five groups (A–E). Each group attends one specialised hall per day and rotates daily so that by Friday, all groups have completed learning in every hall.

STEM Halls and Focus Areas:

- Newton Hall – Electronics & Automation
- Turing Hall – Programming, Software & FinTech
- Wright Brothers Hall – Aerospace & Drone Systems
- Faraday Hall – Energy, Power & Control Systems
- Curie Hall – Medical, BioTech & AgroTechnology

Monday to Friday Hall Rotation Schedule

Day	Group A	Group B	Group C	Group D	Group E
Monday	Newton Hall	Turing Hall	Wright Brothers Hall	Faraday Hall	Curie Hall
Tuesday	Turing Hall	Wright Brothers Hall	Faraday Hall	Curie Hall	Newton Hall
Wednesday	Wright Brothers Hall	Faraday Hall	Curie Hall	Newton Hall	Turing Hall
Thursday	Faraday Hall	Curie Hall	Newton Hall	Turing Hall	Wright Brothers Hall
Friday	Curie Hall	Newton Hall	Turing Hall	Wright Brothers Hall	Faraday Hall

Key Notes

- All groups follow identical daily time blocks within their assigned halls.
- Friday evening is reserved for Technical Reports Presentation and Defense.
- Robotics & Automation performance in Newton Hall informs selection of the Singapore / Switzerland World Robotics Challenge Team.
- Saturday features Exhibitions and the Awards Ceremony.
- Sunday marks official departure.

CAMP PROJECTS

(Build in 7 days)

LESSONS CAMPERS WILL FIND:

A.

PROJECT 1 TO 5 IN SINGLE CLASS.
PROJECT 6 TO 10 IN SINGLE CLASS.
PROJECT 10 TO 15 IN SINGLE CLASS
PROJECT 15 TO 20 IN SINGLE CLASS.

Learners alternate building the projects as a morning session booster during lessons period.



1. Dry–Wet Waste Segregator System



Purpose

To design an automated waste segregation system that separate wet and dry waste, addressing challenges in waste management, recycling, and environmental sustainability.

Real-Life & Industry Application

- Municipal solid waste management
- Smart cities and environmental engineering
- Recycling plants and landfill efficiency systems

Concept Explanation

The system uses a soil moisture sensor to detect moisture content in waste and a servo motor to divert waste into the appropriate bin. An ultrasonic sensor detects object presence to trigger the sorting mechanism automatically.

Learning Outcomes

- Sensor integration and calibration
- Automation logic and conditional programming
- Environmental engineering concepts



2. LED Chaser Circuit



Purpose

To understand digital outputs, timing logic, and sequential control, which are foundational in electronics and industrial signaling systems.

Real-Life & Industry Application

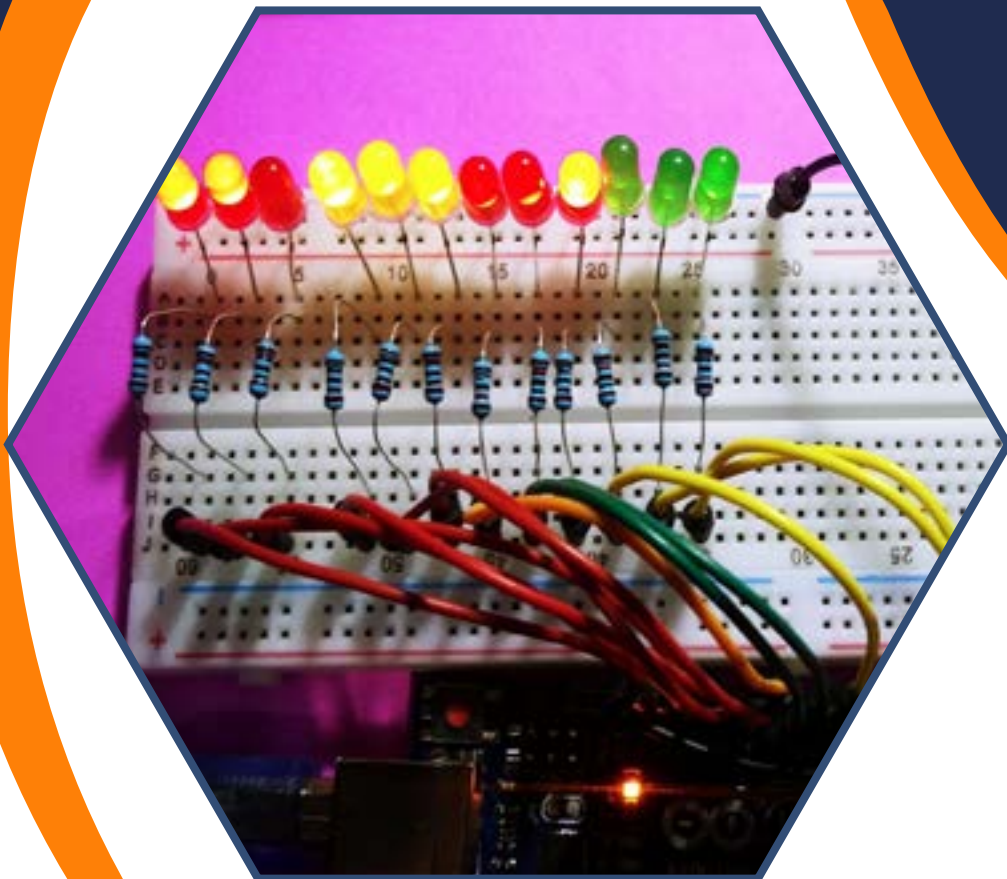
- Industrial status indicators
- Traffic signaling systems
- Decorative and safety lighting systems

Concept Explanation

Multiple LEDs are programmed to turn ON and OFF in a sequence using Arduino IDE, teaching timing control through `delay()` and loop logic.

Learning Outcomes

- Digital pin control
- Basic electronics and resistor usage
- Embedded programming logic



3. Obstacle Avoiding Robot



Purpose

To design an autonomous robotic system capable of detecting and avoiding obstacles, introducing learners to autonomous navigation.

Real-Life & Industry Application

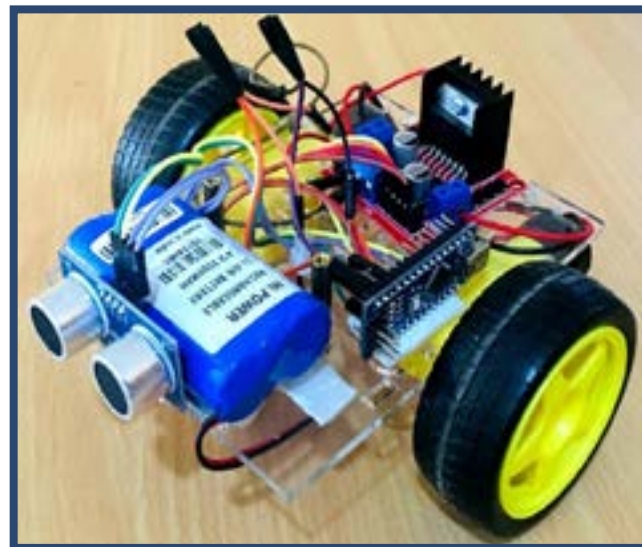
- Autonomous vehicles
- Warehouse robots (AGVs)
- Defense and exploration robots

Concept Explanation

An ultrasonic sensor mounted on a servo motor scans the environment. Based on distance measurements, the robot decides movement using DC motors controlled via a motor shield.

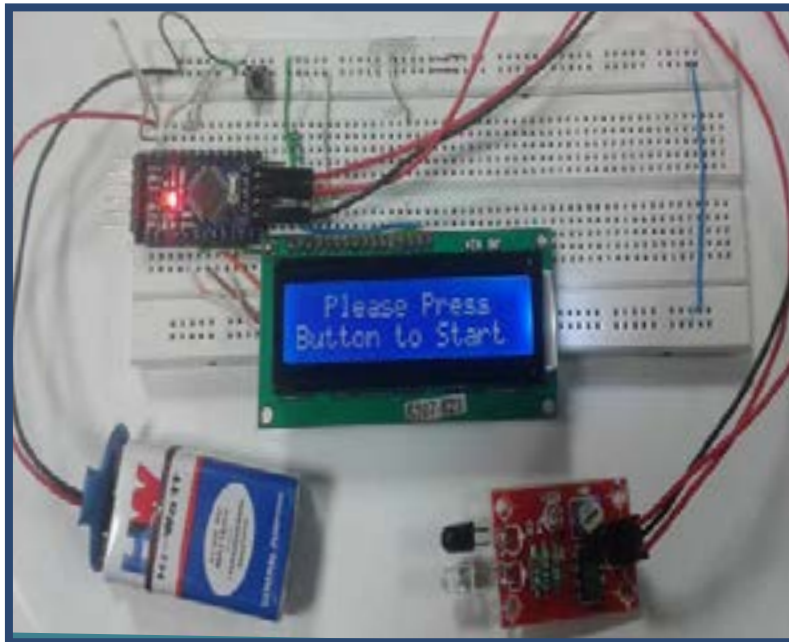
Learning Outcomes

- Robotics fundamentals
- Decision-making algorithm
- Motor control and sensors



4.

RPM Counter (Speed Measurement System)



Purpose

To measure the rotational speed of a motor, an essential concept in mechanical, automotive and manufacturing industries.

Real-Life & Industry Application

- Engine diagnostics
- Industrial machinery monitoring
- Power tools and turbines

Concept Explanation

An IR sensor detects interruptions from a rotating black disc, calculates revolutions per minute (RPM), and displays results on an LCD.

Learning Outcomes

- Speed calculation logic
- Sensor-based measurement
- Data display and instrumentation



5. Automatic Solar Tracker



Purpose

To improve solar panel efficiency by tracking sunlight automatically.

Real-Life & Industry Application

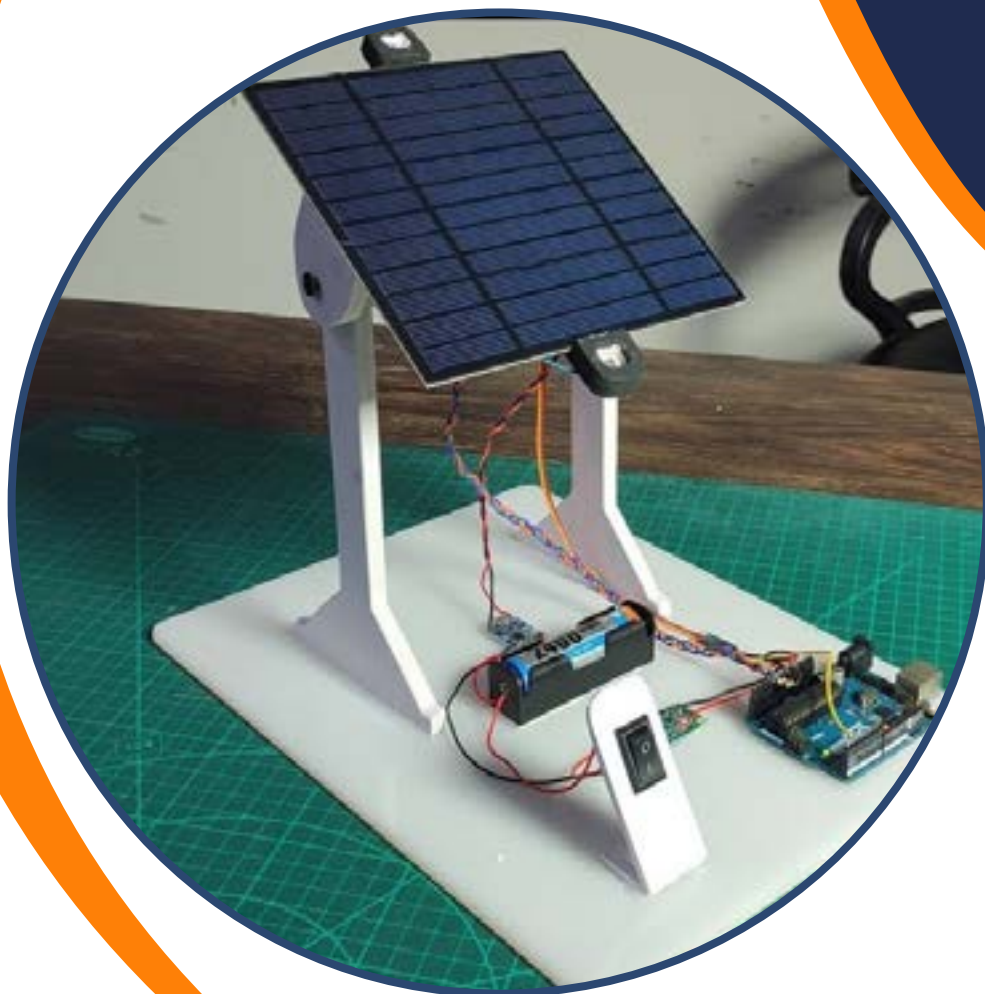
- Renewable energy systems
 - Solar farms
- Smart energy solutions

Concept Explanation

Two LDR sensors compare light intensity and rotate a solar panel using a servo motor to face the direction of maximum sunlight.

Learning Outcomes

- Renewable energy concepts
 - Analog sensor readings
- Energy efficiency optimization



6. DIY Maze Game (Mechatronics & Control)



Purpose

To introduce human-machine interaction, control systems, and precision movement.

Real-Life & Industry Application

- C N C machines gaming hardware design

Concept Explanation

A joystick controls servo motors that tilt a platform, guiding a ball through a maze.

Learning Outcomes

- Joystick input mapping
- Servo motor control
- Mechatronics fundamentals



7. Laser Security Alarm System



Purpose

To design a basic intrusion detection system.

Real-Life & Industry Application

- Home and commercial security
- Military perimeter systems
- Automated surveillance

Concept Explanation

A laser beam aimed at an LDR triggers an alarm when interrupted, simulating real security systems.

Learning Outcomes

- Security system logic
- Light-based sensing
- Alarm and alert mechanisms



8. Automatic Water Level Alarm



Purpose

To prevent water overflow and wastage using automation.

Real-Life & Industry Application

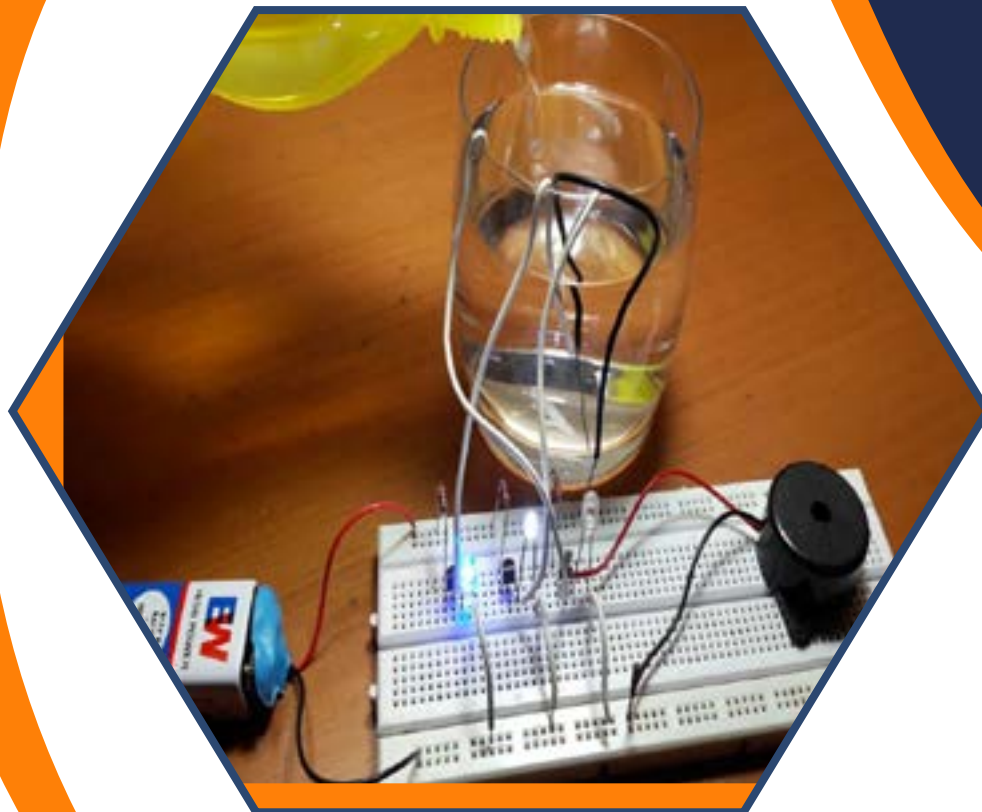
- Domestic water tanks
- Industrial reservoirs
- Smart water management systems

Concept Explanation

An ultrasonic sensor measures water level and triggers alarms while displaying levels on an LCD.

Learning Outcomes

- Distance measurement
- Resource conservation
- Display systems



9. Smart Door Lock Using RFID



Purpose

To introduce access control and authentication systems.

Real-Life & Industry Application

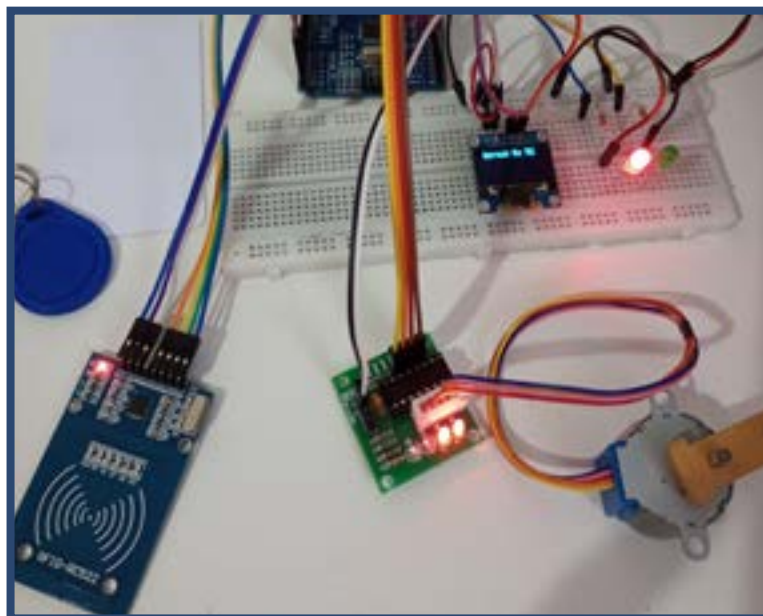
- Smart homes
- Office access systems
- Hotel keycard systems

Concept Explanation

An RFID reader authenticates cards and controls a servo motor lock while providing buzzer alerts.

Learning Outcomes

- RFID technology
- Embedded security concepts
- Authentication logic



10. Automatic Cloth Drying Stand



Purpose

To protect clothes from rain using environmental sensing.

Real-Life & Industry Application

- Smart homes
- Weather-responsive automation

Concept Explanation

A rain sensor detects rainfall and triggers a servo to retract clothes automatically.

Learning Outcomes

- Environmental sensors
- Home automation
- Decision-based programming



11. DIY Robotic Arm



Purpose

To simulate industrial robotic manipulators.

Real-Life & Industry Application

- Manufacturing automation
 - Assembly lines
 - Medical robotics

Concept Explanation

Joystick-controlled servo motors move a robotic arm built from simple materials.

Learning Outcomes

- Robotics kinematics
 - Control systems
 - Mechanical creativity



12. Reverse Car Warning System



Purpose

To enhance vehicle safety.

Real-Life & Industry Application

- Automotive parking systems
- Driver assistance systems

Concept Explanation

An ultrasonic sensor detects obstacles and triggers LEDs and buzzers with varying intensity.

Learning Outcomes

- Automotive electronics
- Safety system logic



13. Dino Game (Embedded Gaming)



Purpose

To teach logic design and display control through game development.

Real-Life & Industry Application

- Embedded user interfaces
- Game development logic

Concept Explanation

A simplified version of the Dino game runs on an LCD using button inputs.

Learning Outcomes

- State machines
- Display programming



14. Bluetooth-Controlled Robot



Purpose

To introduce wireless control systems.

Real-Life & Industry Application

- Remote robotics
- IoT-based mobility

Concept Explanation

A smartphone communicates with the robot via Bluetooth to control movement.

Learning Outcomes

- Wireless communication
- Human-robot interaction



15. Line Following Robot



Purpose

To implement automation based on feedback systems.

Real-Life & Industry Application

- Factory conveyor systems
- Automated logistics

Concept Explanation

IR sensors detect a line and guide the robot accordingly.

Learning Outcomes

- Feedback control
- Sensor fusion



16. Digital Dice



Purpose

To teach random number generation and display logic.

Real-Life & Industry Application

- Embedded entertainment systems

Concept Explanation

X

Learning Outcomes

- Digital displays
- Randomization logic



17. DIY Sonar System



Purpose

To visualize distance sensing similar to radar and sonar technologies.

Real-Life & Industry Application

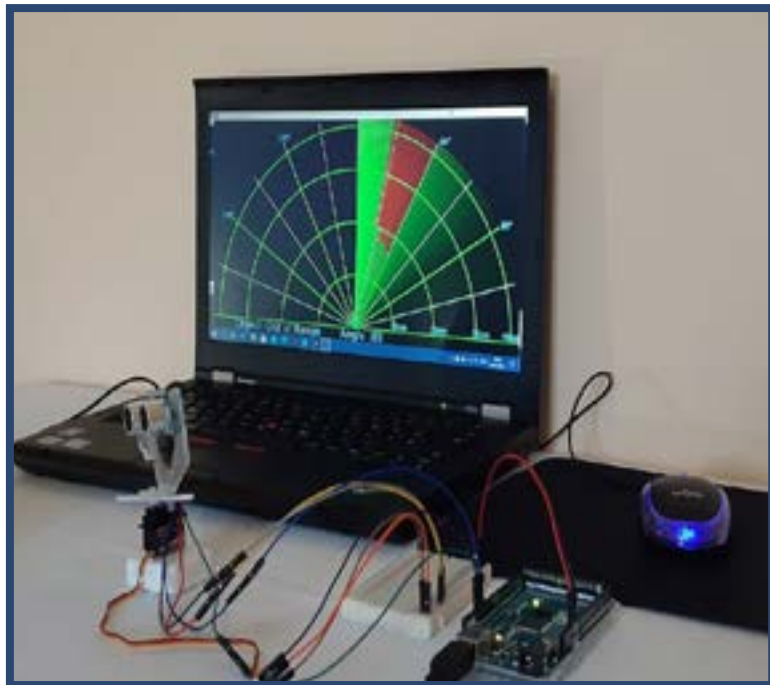
- Marine navigation
- Defense systems

Concept Explanation

X

Learning Outcomes

X



18. Number Showcasing System



Purpose

To understand numeric representation in embedded systems.

Real-Life & Industry Application

- Digital meters
- Counters

Concept Explanation

X

Learning Outcomes

X



19. Blind Stick (Assistive Technology)



Purpose

To support visually impaired individuals through technology.

Real-Life & Industry Application

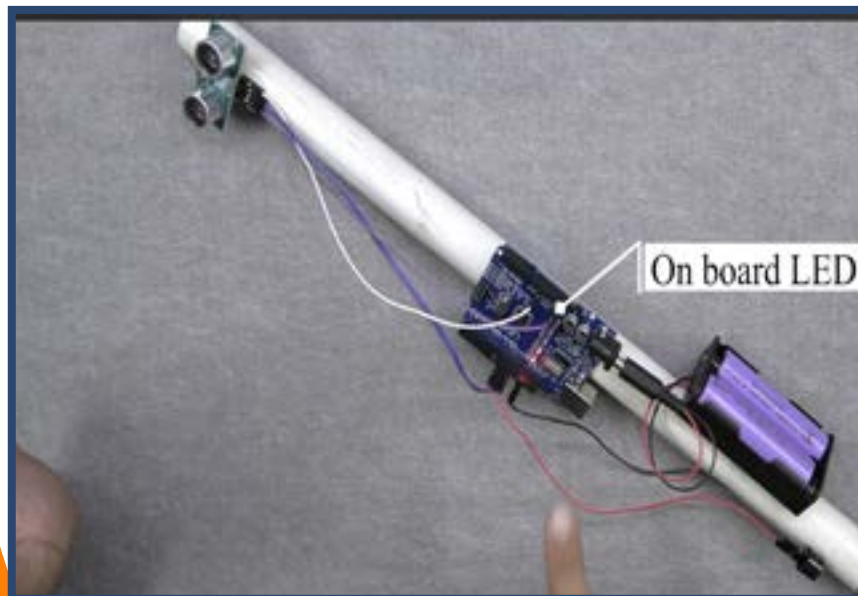
- Assistive devices
- Healthcare engineering

Concept Explanation

X

Learning Outcomes

X



20. Smart Dustbin



Purpose

To improve hygiene and sanitation using automation.

Real-Life & Industry Application

- Smart cities
- Public sanitation systems



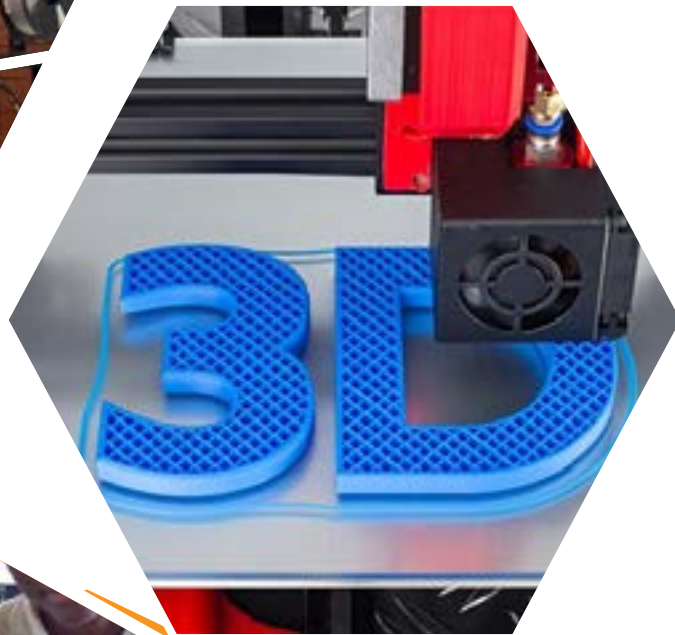
Final Notes On Above Projects.

These projects collectively expose campers to:

- Automation
- Robotics
- Embedded systems
- Problem-solving for real societal and industrial challenges



B. 3D PRINTING SECTION



One Classroom will be dedicated to House 10(ten) 3D Printers 9(nine) Assembled, 1(one) will be used to teach the learners the process of Assembling and making the 3D Printer Ready For Usage.

Computers with SOLID-DWORKS SOFTWARE and others will be available for usage in creating 3D models by University students who would want to practice more having foundation in CAD Design.

For all other learners designs will be made available for printing and fundamentals of 3D printing will be taught in this class.



C. DRONE FLYING CLASS



A Theory class will be held for 1 hour on fundamentals of flight explaining the basic physics of flight. After which building/ Assembling of Drones to understand fundamental components and electronic wiring. Flight practice will be done and the Assembled drones will be used along side the Promabridge fleet.

Componets



Assembly



Flight

After assembly flight lessons will commence, learners would be flying drones with complete understanding of which component performs a specific function.

PROJECT AREAS FOR CAMPERS (Build in 5 Days)

Projects will include, but not limited to:

- Hydroelectric Power Plant Design (Form 4 to University) Requirements
- Drone Manufacturing (Form 1 to University) Requirements
- Smart Agriculture Monitoring Systems Requirements
- Robotics Building Competition (Grade 4 to University) Requirements
- Website Development for E-Commerce using HTML or Python (Form 2 to University) Requirements
- Automatic Control Systems (Open to all)
- National Security Digital Solutions (University Level)
- Mining Technology Solutions
- Medical Technology Innovations
- Petrochemical Industry Projects

Detailed project sketches, capacity requirements, and technical documentation are under development and will be shared with the committee. (Materials required per project are under review and mentors to skilled in each field will be selected among experts and Promabridges STEM Educators) This proposal gives an indepth cover of the camps expectations and requirements but further budgeting of components and equipments required to deliver the quality put in the planning of this STEM Camp is under review by the office of the Chief Financial officer Promabridge)

Expected Expert Sessions



PACRA | Company Registration and Patent Filing



ZRA | Education department
(Tax Compliance as young Business owner)



ZDA | Zambia Development Agency.
(Opportunities For Youths In Business)

Ministerial Presence:

1. Minister of Science and Technology
2. Ministry of Education



Sponsors are invited to partner in building Zambia's innovation future through practical STEM education.

SPONSORSHIP PACKAGE PRICING



PREMIUM PACKAGE (Strategic Partner)

PRICE: ZMW 250,000 – 350,000

WHO THIS IS FIT FOR:

- Major banks (Zanaco, Absa, Stanchart)
- Mining companies
- Large tech or telecom-linked institutions
- Government agencies as strategic partners (often in-kind)

WHY THIS PRICE? based off research:

- Banks in Zambia regularly spend K500k+ annually on CSR
- This gives them naming rights + policy alignment + visibility



GOLD PACKAGE (Program Partner)

PRICE: ZMW 120,000 – 180,000

WHO THIS IS FIT FOR:

- Second-tier banks.
- Insurance companies.
- Mid-size corporates.
- Education institutions

WHY THIS PRICE WOULD WORK:

- Affordable CSR spend.
- Sponsors one major camp pillar (finance, robotics, entrepreneurship)



SILVER PACKAGE (Supporting Partner)

PRICE: ZMW 40,000 – 75,000

WHO THIS IS FIT FOR:

- SMEs.
- Individuals of influence.
- Education providers.
- NGOs & foundations

It's attractive WHY ?:

- -Entry point.
- -Visibility + goodwill.
- -No heavy commitments.



Conclusion

Zambia stands at a defining moment in its development journey. The PromaBridge National STEM Innovation Camp 2026 is more than a holiday program, it is a national investment in practical skills, innovation capacity, and future economic strength.

By equipping young people with hands-on STEM experience, entrepreneurial exposure, and industry engagement, this initiative bridges the gap between education and real-world problem solving. It transforms learners into creators, thinkers, and builders prepared to contribute meaningfully to national growth.

An investment in practical STEM education today secures Zambia's competitiveness tomorrow. Through strategic partnerships and committed sponsorship, we can cultivate a generation capable of driving industrial growth, technological advancement, and sustainable development by 2050.

Together, we build innovators.
Together, we build capacity.
Together, we build Zambia.

SOME ADD-ONS, WE HOPE TO BE SEEN
BENEFICIAL TO THE CAUSE, just for flexibility:

- In-kind sponsorships(equipment, internet, training): valued & credited.
- Media sponsorship: ZMW 30,000 – 50,000 equivalent.
- Transport sponsorship: ZMW 20,000 – 40,000

Meet Our Team



CASTROL WASWAVA
MZYOOMA

“Mechanical systems expert in IoT, robotics, and automation with 3D printing skills, Python/C++ proficiency, and entrepreneurship experience at Marwadi University.”



WISDOM SIMWILA
KALUNGA

“Engineer skilled in jet engine design, aeroacoustic, and propeller innovation, driving PromaBridge's mission to turn student ideas into real-world solutions.”



MAUREEN CHIBANGA
MBChB

“Medical expert and STEM advocate dedicated to mentoring future Doctors, Nurses, Lab Technicians, and Scientists, with hands-on experience in advanced medical procedures.”



DR, HAPPY SIAME

“A mathematician with a PhD, specializes in modeling, simulations, and algorithms, applying Python/C++ to advance AI, data science, and STEM education.”



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